

Engg. - 46
T.E. (Computer Engg.) (Semester - I)
THEORY OF COMPUTATION (In Sem.)
(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

Unit - I

- Q1)** a) For given relations $R_1 = \{(1, 2), (2, 3), (2, 4)\}$ and $R_2 = \{(1, 2), (2, 3), (2, 4)\}$ Find R^+ . [4]
 b) Is NFA more powerful than DFA? Justify. [2]
 c) Construct a DFA accepting all strings w , over $\{0, 1\}$ such that the number of '1's in w is multiples of 3. [4]

OR

- Q2)** a) Write regular expressions for each of the following languages : [6]
 i) For $\Sigma = \{a, b\}$, set of all strings with no consecutive 'a' s and 'b' s
 ii) For $\Sigma = \{0, 1\}$, set of all strings in which every 0 is immediately followed by at least two 1's.
 iii) $L = \{1^{(2n+1)} \mid n \geq 1\}$
 b) Define : [4]
 i) Symbol ii) String
 iii) L^* and iv) L^+

Unit - II

- Q3)** a) Construct a Moore machine equivalent to the Mealy machine M defined in the following table with input alphabet, $\Sigma = \{0, 1\}$. [6]

Present state	Next state			
	$\Sigma = 0$		$\Sigma = 1$	
	State	output	state	output
-> q_1	q_1	1	q_2	0
q_2	q_4	1	q_4	1
q_3	q_2	1	q_3	1
q_4	q_3	0	q_1	1

P.T.O.

- b) Construct NFA without ϵ moves from given NFA with ϵ moves. [4]



OR

- Q4)** a) Apply the theorem of Pumping lemma to prove that the following language L is regular. $L = \{a^{2n} | n \geq 1\}$. Draw the finite automata for L. [6]
- b) State applications for Regular Expressions. [4]

Unit - III

- Q5)** a) Convert the following right-linear grammar to its equivalent left-linear form. [6]
- $S \rightarrow 0A \mid 1B$
 $A \rightarrow 0C \mid 1A \mid 0$
 $B \rightarrow 1B \mid 1A \mid 1$
 $C \rightarrow 0 \mid 0A$
- b) Explain with suitable examples the Chomsky Hierarchy for classification of languages. [4]

OR

- Q6)** a) Convert the following grammar to Chomsky Normal form (CNF) [6]
- $S \rightarrow bA \mid aB$
 $A \rightarrow bAA \mid aS \mid a$
 $B \rightarrow aBB \mid bS \mid b$
- b) Give context free grammars for the following languages. [4]
- i) $L = \{x \mid x \in \{a, b\}^* \text{ with strings of starting with 'a' and ending in 'b'}\}$
 ii) $L = \{x \mid x \in \{a, b\}^* \text{ with strings of even length palindrome}\}$
